# GoUp AI Performance Marketing Intelligence Agent: Architecture (Phase 3)

Oct 13, 2026 · @Ommala deiii

## 1. Purpose, scope and status labels

This document proposes the architecture for the approved MVP: W1 launch readiness, W5 diagnosis and W6 scale decision, plus the shared foundations they need. It is for review. It contains no application code and does not reopen any Phase 2 or Phase 2.1 decision.

**What I read.** The Phase 1 Knowledge Blueprint, the Phase 2 Product Specification, the Phase 2.1 Specification Addendum, and the GoUp toolkit PDF. The Phase 2 review handoff was not attached as a separate file, so I treated the Phase 2.1 Addendum (dated 13 Oct 2026) as that handoff. Where the Addendum and the toolkit differ, the toolkit wins.

**Status labels.** Every rule keeps the label it already has. I add one label for engineering choices so they are never mistaken for method.

| Label | Meaning | Source |
|---|---|---|
| GoUp | Stated in the toolkit, page cited. Applied as written | Toolkit |
| Decision | Product choice by the product owner. Fixed unless reversed | Phase 2, Phase 2.1 |
| Provisional | Proposed to remove an ambiguity. Not GoUp method. Runs as a configurable default. Outputs say "Provisional, not GoUp method" | Phase 2.1; new items here carry an ID like P3-01 |
| Marketer-set | Defined per client by the marketer. Never presented as GoUp | Client context |
| Architecture | An engineering choice that decides how the product is built. Changes no rule and no verdict, and can be changed without GoUp involvement | This document |

Any Architecture choice that would change a visible behaviour is not labelled Architecture. It is raised as Provisional with an ID in section 12.

**Out of scope for this document.** Technology and vendor selection, UI design, prompt text, the deferred workflows (W2, W3, W4, W7), screenshot extraction, retention and referral logic.

## 2. What the architecture must guarantee

Ten guarantees come straight from Phase 2 and 2.1. Each one fixes a structural choice, so they are listed first and the components follow from them.

| # | Guarantee | Source | What it forces |
|---|---|---|---|
| G1 | Every number, gate, verdict, level and step comes from a deterministic engine. The model never calculates or decides | Addendum §5 (Decision) | A pure engine with no model access. Narrative numbers are bound to engine figures |
| G2 | Every rule shows its status and strength wherever it is relied on | Addendum §1, Spec rule 12 | Rules live in a registry. Labels are added by code, not written by the model |
| G3 | A failed gate is never relabelled. Overrides are logged next to the verdict | Spec §10, Addendum §6 | Verdicts are immutable. An override is a separate linked record |
| G4 | Known, assumed and unknown stay apart | Spec rule 5 | The split is computed from field status and evidence flags, not from model judgement |
| G5 | Evidence is confirmed by the user before use | Addendum §2, §5.1 | Confirmation is a UI action the model cannot call |
| G6 | A stored decision re-runs to the same verdict | Addendum §5.9, §6, §7 | Engine is a pure function of versioned inputs. Time is an input, not a clock |
| G7 | One client per session, no cross-client reads | Addendum §6 (Decision) | Scope is bound at session start and enforced in storage, not only in prompts |
| G8 | No credentials. Pasted content cannot give instructions | Addendum §6, §7 layer 4 | Input scanner before the model. Extraction runs with no tools |

| # | Guarantee | Source | What it forces |
|---|---|---|---|
| G9 | Provisional rules can change without invalidating the whole test set | Addendum §7 | Provisional rules are versioned configuration with register IDs |
| G10 | The human decides. The agent never executes or touches ad accounts | Spec §2, rule 11 | No write path to Meta. The model's tools can only propose or read |

Two further constraints shape the build. Hard delete on request must coexist with append-only history (section 10). And GoUp's own method is thin in places the engine must still decide, so section 12 lists every new Provisional rule this design needs.

## 4. Components and boundaries

Eleven components, grouped by what is allowed to be wrong. [Architecture] For the MVP they ship as one deployable application with hard module boundaries, not as separate services. The engine is the one part built as a standalone library with no I/O so it can be tested, replayed and reused without the rest.

| Component | Zone | Responsibility | Must never |
|---|---|---|---|
| Session gateway | Deterministic | Authenticates the marketer, binds the session to one client, scans inbound text for credentials, enforces scope on every call | Let a session read or write another client |
| Orchestrator | Deterministic | Runs the W1, W5 and W6 state machines. Uses the field matrix (B and C) to decide the next missing field and when to call the engine | Interpret free text or compute |
| Model layer | Model-assisted | Three roles with different permissions: extractor, dialogue and hypotheses, narrative writer (section 9) | Calculate, round, decide a gate, confirm anything |

| Component | Zone | Responsibility | Must never |
|---|---|---|---|
| Evidence intake | Deterministic, model-assisted mapping | Parses typed entries, pasted tables and CSV into staged items. Asks the user to confirm each value, unit, currency and period | Pass an unconfirmed or image-derived value to the engine |
| Client context service | Deterministic | Stores context fields with status, source, confirmer, time and version. Detects staleness and conflicts. Produces snapshots | Overwrite silently. History is append-only |
| Rule registry | Deterministic | Holds every rule, definition slot, checklist item, ladder and parameter as versioned data with status, strength and page | Be edited at runtime by a session or the model |
| Engine | Deterministic | Pure functions: metrics, guardrail position, sufficiency, tracking suggestion, staleness, W1 gate, W5 state and ladder lookup, W6 sequence, verdict mapping, Reduce rule, step and review window | Call the model, read the clock, touch storage |
| Output gate | Deterministic | Checks the narrative against the engine result, binds numbers, adds status and strength labels, builds the standard response shape | Release text with an unbound number or a missing label |

| Component | Zone | Responsibility | Must never |
|---|---|---|---|
| Decision ledger | Deterministic | Stores launch, learning and scaling records with every version needed to replay them. Drafts are model-assisted, commits are human | Edit a committed verdict |
| Governance | Deterministic | Retention setting, per-client export, hard delete, rule configuration releases, register upkeep | Share data across clients |
| Evaluation and replay harness | Offline and runtime | Runs the seven evaluation layers and re-runs stored decisions (section 11) | Be skipped before a release |

**Boundary rule.** Data flows into the engine only through evidence items and context fields that carry a confirmed flag. Data flows out of the engine only as structured results. The model sees results by figure ID and rule ID, never as raw tables it could re-derive.

## 5. Rule registry and knowledge layer

The MVP's knowledge is small, enumerable and mostly rule-shaped, so it is stored as typed, versioned data and looked up by ID. It is not stored as documents for semantic retrieval. [Architecture] Two reasons: retrieval can drop the strength label or blend a worked example into a rule, and the fidelity test (layer 2) needs a closed set of statements to check. Page numbers below are PDF page numbers, as in the Blueprint.

**What a registry entry carries.** Every entry has an ID, a kind, a status (GoUp, Decision, Provisional, Marketer-set), a strength (firm rule, guideline, illustrative, provisional), the page cited, a short statement in GoUp's words, parameters, the workflows it applies to, a register ID if Provisional, and a version. A Provisional entry also carries its default and whether the marketer can override it.

**Seven kinds of entry**, taken from the Blueprint's split of knowledge by type:

| Kind | Treated as | MVP examples |
|---|---|---|
| Gate | Evaluated by the engine, yields Pass, Fail or Cannot evaluate | Launch Go/No-Go; the seven scaling questions |
| Ladder | Candidate generator for hypotheses, never a verdict | Funnel diagnostic ladder, creative diagnosis, lead-quality diagnosis |
| Catalogue | A list the marketer ticks through | Launch checklist items, tracking audit items, capacity checks |
| Calculation | Fixed formula, labelled GoUp or standard arithmetic | Marginal cost per outcome; CPL; break-even ROAS |
| Guideline range | Offered as a soft starting point, client guardrails override | Budget levels and steps, review windows |
| Definition slot | A term GoUp uses but never defines. Filled by the marketer | Reasonably stable, acceptable range, minimum evidence, critical |
| Worked example | Test fixture only. Never reachable by the model as a default | Campaign A and B, the 300 to 360 guardrail |

**Seed content and its labels.** Pages marked verified were checked against the toolkit PDF for this document. The rest are carried from the Blueprint and Addendum and are checked at seeding time (section 13).

| Family | Status and strength | Pages |
|---|---|---|
| Launch Go/No-Go: Go needs goal, offer, funnel, tracking, creative and sales process ready; No-Go on any unresolved critical conversion-path or tracking issue | GoUp, firm rule | p.5, verified |
| Launch checklist items, eight sections, plus launch baseline capture | GoUp, firm rule | pp.2–5, verified |
| No launch date until access, assets, offer, tracking and client duties are confirmed | GoUp, firm rule | p.41, verified |
| Tracking audit, ten items; High, Medium, Low named with no criteria | GoUp for items. Levels are GoUp labels only | p.34, verified |

| Family | Status and strength | Pages |
|---|---|---|
| "Do not aggressively scale when tracking is unreliable"; business data should roughly match | GoUp, firm rule | p.54, verified |
| Seven scaling questions, in order | GoUp, firm rule | pp.75–76, verified |
| Scale, Hold, Reduce, Diagnose definitions | GoUp | p.62, verified |
| Budget levels L1 to L4 and step ranges | GoUp, guideline ("practical guidelines, not fixed rules") | pp.56–59, verified for L1 and the disclaimer |
| Review windows 3–7 and 14–30 days | GoUp, guideline | pp.74–75, verified |
| Guardrail structure (target, acceptable range, minimum quality rate) | GoUp, firm rule. Values are Marketer-set | p.74, verified |
| Marginal performance method | GoUp | pp.73–74 |
| Symptom-to-suspect ladders | GoUp | T2 pp.12–13, T4 pp.28–29, T5 section 9 |
| Business-type funnel, primary metric, lever order | GoUp | pp.64–68, 78–80 |
| Scaling Log fields | GoUp. "Test" becomes "Next test" (Decision) | pp.77–78 |
| Scale, Hold, Reduce, Diagnose as the only verdicts; seven questions as the single sequence; budgets per day unless stated | Decision | Phase 2 section 4 |
| Tracking policy principle; constrained Scale principle; deterministic engine; one client per session | Decision | Addendum sections 3, 5, 6 |

| Family | Status and strength | Pages |
|---|---|---|
| Tracking criteria, critical items, verdict mapping, Reduce rule, constrained Scale conditions, step boundaries, staleness, retention | Provisional | section 12 |

**Outside the registry on purpose.**

- Meta platform behaviour is volatile and not GoUp method. The model may mention it only as a typed platform claim, and the output gate attaches "verify in Meta's current documentation".
- General marketing knowledge is allowed only as a typed general-knowledge claim, labelled "general knowledge, not GoUp method". It can never feed a gate, score or verdict.
- The stray note at the end of p.5 is excluded from all content (Phase 2 decision).

**Release discipline.** The registry changes only through a versioned release. A release publishes the rule configuration version and runs the regression set. A session reads one pinned version, so a rule change never alters a half-finished conversation.

## 6. Deterministic engine

The engine is a pure function: confirmed inputs, a pinned rule configuration and an as-of time go in, and one structured result comes out. [Architecture] It has no I/O, no model call and no clock, uses decimal arithmetic, and is total, meaning it always returns a result, even when that result is "cannot evaluate". This is what makes G1 and G6 testable.

**Engine modules.**

- Calculation: the metric dictionary in Addendum §5, with zero-denominator, no-incremental-outcome, spend-decrease and impossible-input handling.
- Reference and sufficiency: resolves the baseline, target or prior period, and states each result against the marketer's minimum-evidence rule.
- Guardrail position: within target, within range or beyond range, compared before rounding, bounds inclusive (PV-08).
- Tracking assessor: suggests High, Medium or Low from the ten audit items and reconciliation (PV-01), applies the new-campaign and staleness caps.
- Staleness: marks fields stale from change events and the marketer's interval (PV-03).
- W1 gate: Go or No-Go from the launch checklist, the critical set and the tracking level.

- W5 classifier and ladder lookup: states each funnel stage healthy, weak or cannot conclude, then returns candidate suspects from the ladders.
- W6 sequence: the seven questions, verdict mapping, Reduce rule, constrained Scale safeguards, lever order, step and review window.

**Figure contract.** Every figure returns a value, formula ID, inputs, period, unit or currency, a status (computed, derived, not computable, no incremental outcomes, below minimum evidence) and flags. Display rounding is stated and never changes a comparison. Narrative may quote only these figures.

**Gate and result states.** A question returns Pass, Fail, Cannot evaluate, or at Q1 only, Conditional (Medium tracking). Every non-Pass state names the field or reason behind it. The engine also tags each input as known (user-confirmed), assumed (derived, default or unconfirmed) or unknown (missing), which is how the response's known, assumed and not yet known sections are produced.

**W6 evaluation order.** The Addendum's table maps each question to a verdict but leaves four cases open. This is how I propose the engine closes them. Each is a Provisional rule in section 12.

1. Apply caps to the confirmed tracking level: a new campaign cannot be High, and a stale confirmation counts as Medium at most (PV-01, PV-03).
2. Q1 from the level. High is Pass. Low is Fail, verdict Diagnose with action "fix tracking". Medium is Conditional: the sequence continues so the safeguards can be tested, and "resolve the tracking uncertainty" stays attached to whatever verdict results (PV-11).
3. Q2 to Q6 in order. The first Fail ends the sequence with its mapped verdict. The first Cannot evaluate ends it with no verdict, names the missing field and why it matters, and lists the later questions as not evaluated. A separate readiness preview lists every blocking field still missing for later questions, so the marketer can supply them in one pass (PV-12).
4. At Q2, Pass or Fail comes from the primary KPI against the client's guardrail only. Evidence sufficiency does not flip it. It blocks Reduce and is what fails Q4 (PV-13). A Q2 Fail becomes Reduce only when all four Reduce conditions hold (PV-06), otherwise Diagnose. Reduce gives direction only, never a step size.
5. If Q1 was Conditional and Q2 to Q6 all pass, test the six safeguards (PV-04). If all hold, the verdict is "Scale, constrained (Medium tracking)". If any fails, it is Diagnose and the failing safeguard is named. Three safeguards rest on marketer confirmations (the open item does not affect how the outcome is measured, the risk is acknowledged, a stop condition and review window are set). They are recorded as confirmations with a reason, not inferred.

6. At Q7 the engine returns the business-type lever order with each lever's preconditions. The marketer confirms one lever, or confirms none is justified, which gives Hold (PV-14). Under Medium tracking only the budget lever is allowed, with the engine's smallest guideline step. If no guideline step exists (non-INR, below ₹300 a day, or L4), safeguard 4 cannot be evaluated and the verdict is Diagnose (PV-18).
7. Attach the step as a labelled guideline, the guardrail range, the review window class, the main risk and the metric to watch, then draft the Scaling Log entry.

**Budget step checks (fixtures).** With the Addendum's boundary rule (PV-07): ₹300 gives L1 and 10%. ₹1,000 gives L1 and 10%. ₹3,000 gives L2 and 10%. A budget strictly inside a level gets that level's lower-end step (₹2,000 gives L2 and 15%). Exactly ₹10,000 resolves to L3 and 10% under the boundary rule even though the Addendum also words L4 as "10,000 and above"; I flag that for confirmation. Above ₹10,000, below ₹300 or any non-INR account gives no percentage and asks the marketer.

**Truth-table size.** Q1 is a function of the tracking level, so the independent space is Q2 to Q6 at three states each, crossed with four tracking states (High, Medium, Low, stale): 243 × 4 = 972 rows before Q7, safeguard and Reduce sub-inputs. That is small enough to test exhaustively.

## 7. Data model

Three rules shape the data model. History is append-only, so a change is a new version and never an overwrite. Every engine run points at an immutable snapshot of the context and a hash of the evidence it used. Every record names the client it belongs to. [Architecture] A relational store with row-level scoping by client fits this; the choice of product is left to implementation.

| Entity | Key fields | Mutability | Notes |
|---|---|---|---|
| Agency | retention setting, active rule configuration | Versioned | Retention is Provisional (PV-09) |
| User | agency, role (marketer, admin) | Versioned | Only marketers operate the agent |
| Client | agency, currency, market, language, business type | Versioned | Currency decides whether rupee guideline values apply |

| Entity | Key fields | Mutability | Notes |
|---|---|---|---|
| Context field | client, key, value, status (confirmed, assumed, unknown), source (user stated, pasted data, agent-proposed and accepted), confirmed by, confirmed at, version, supersedes | Append-only | One row per version. Volatile fields carry a "still true?" flag |
| Definition | client, term (minimum evidence, reasonably stable, acceptable range, critical items, reconciliation tolerance, deviation band, cycle class), structured value, options offered, accepted by | Append-only | Stored as Marketer-set. Options shown are built from toolkit wording |
| Guardrail profile | client, KPI, target, acceptable low and high, minimum quality rate | Append-only | Values are Marketer-set. GoUp's 300 to 360 example can never populate it |
| Tracking record | client, ten audit items with state (working, failing, not checked, not applicable), suggested level, confirmed level, override reason, reconciliation counts and period, campaign age class | Append-only | Level is suggested by the engine, confirmed by the marketer |
| Capacity record | client, sales, inventory, appointments, delivery, support, cash flow | Append-only | Volatile |
| Change event | client, type (site, form, pixel, events, CRM link, offer, creative set, audience, budget, page), date, note | Append-only | Drives staleness and the Reduce rule |

| Entity | Key fields | Mutability | Notes |
|---|---|---|---|
| Launch baseline | client, campaign, metrics, period | Append-only | Captured at launch, used by W5 and W6 |
| Evidence item | client, metric, value, unit, currency, period start and end, entity, origin (typed, pasted, CSV), derived flag, confirmed by, confirmed at, raw reference | Append-only | Only confirmed items are visible to the engine. No image-derived values |
| Engine run | client, workflow, context snapshot ID, evidence hash, input hash, engine version, formula set version, rule configuration version, as-of time, structured result, result hash | Immutable | The unit of replay |
| Response record | engine run, typed claims, rendered text, output gate report | Immutable | Kept for evaluation and audit |
| Decision record | type (launch, learning, scaling), engine run, verdict, qualifier, deciding gate, main risk, review date, next test; later: result, learning, next decision | Draft, then committed. Later fields added as a new version | Replays to the same verdict |
| Override | decision record, gate overridden, stated risk, who, when | Immutable | The verdict is never relabelled |
| Audit event | client, actor, action (confirm value, confirm change, confirm level, commit record, override), time | Immutable | Every human-only action leaves one |

**Snapshot.** A context snapshot is the set of current field versions at the moment of a run, identified by a hash. Replay loads the snapshot and the evidence set, not "the latest".

**Required-field matrix.** The Addendum's B and C matrix is stored as registry data per workflow. The orchestrator reads it to decide what to ask next, so "which field is missing and why it matters" comes from data, not from model wording.

**Learning record.** The Addendum names a learning record from diagnosis. The toolkit defines fields for the Scaling Log (pp.77–78) and the Creative Learning Log (p.13), and has a learning-document prompt (Prompt 18), but no record shaped for diagnosis. The learning record's field set is therefore a Provisional design item (PV-19), built from Prompt 18 and the response shape.